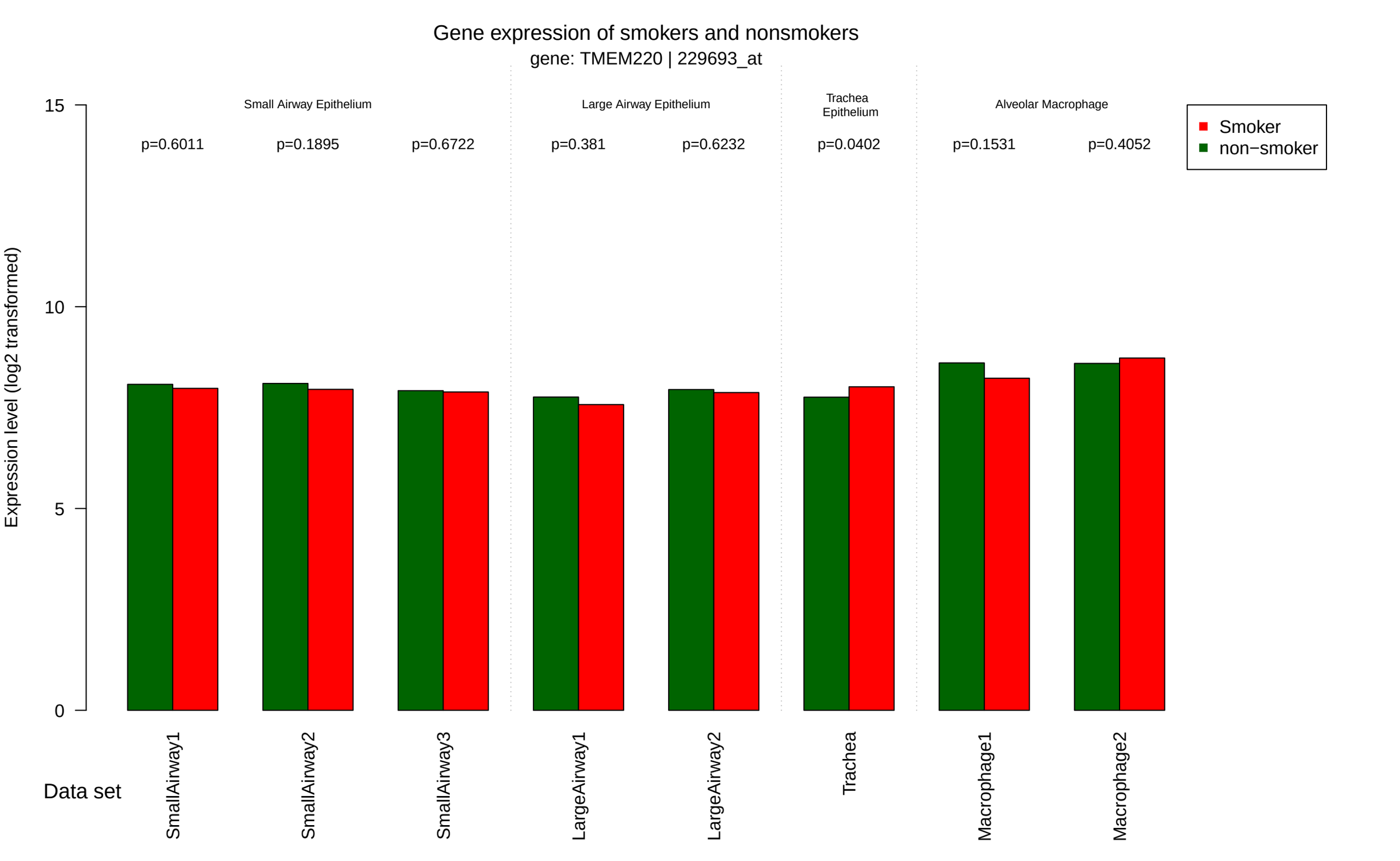
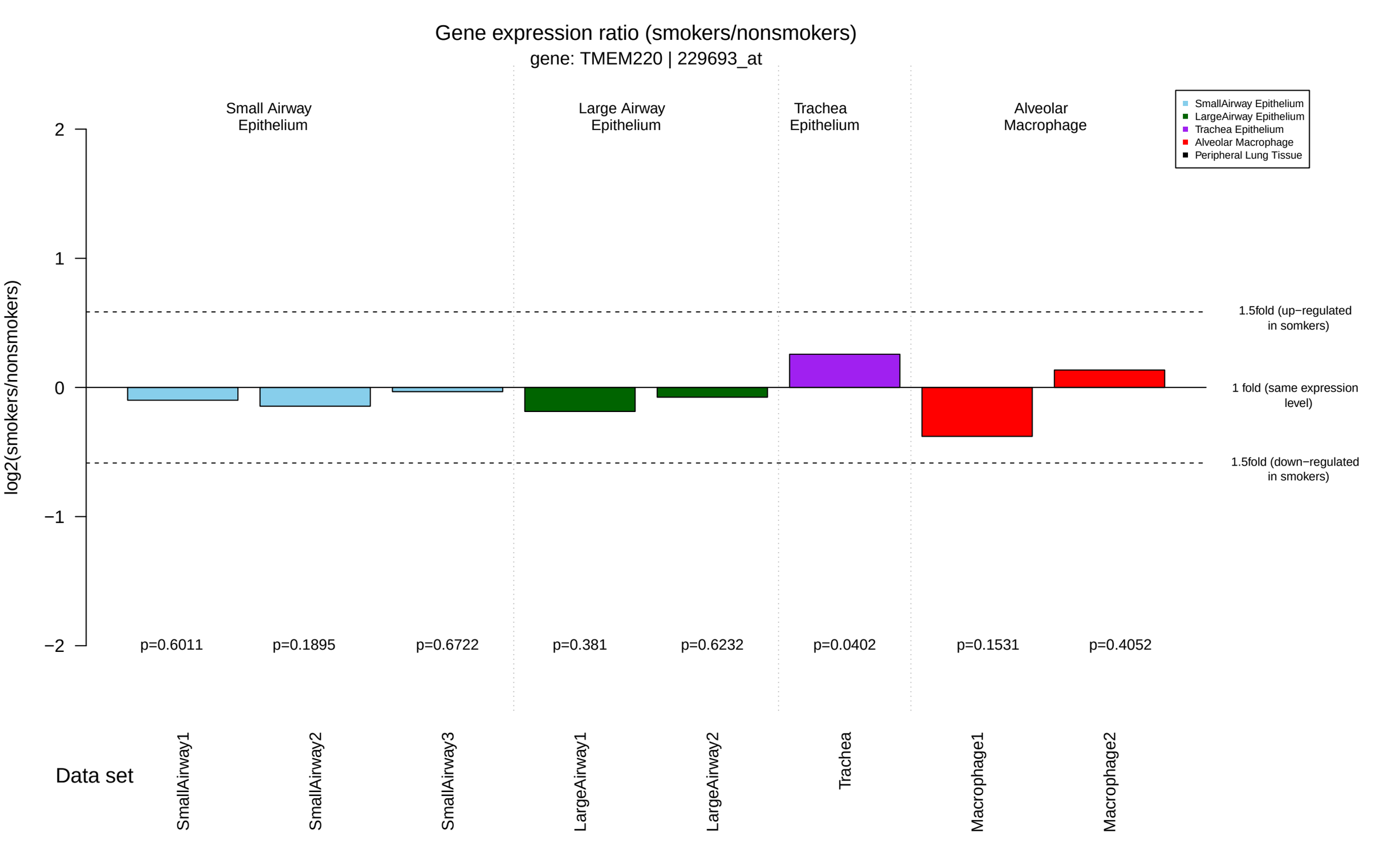
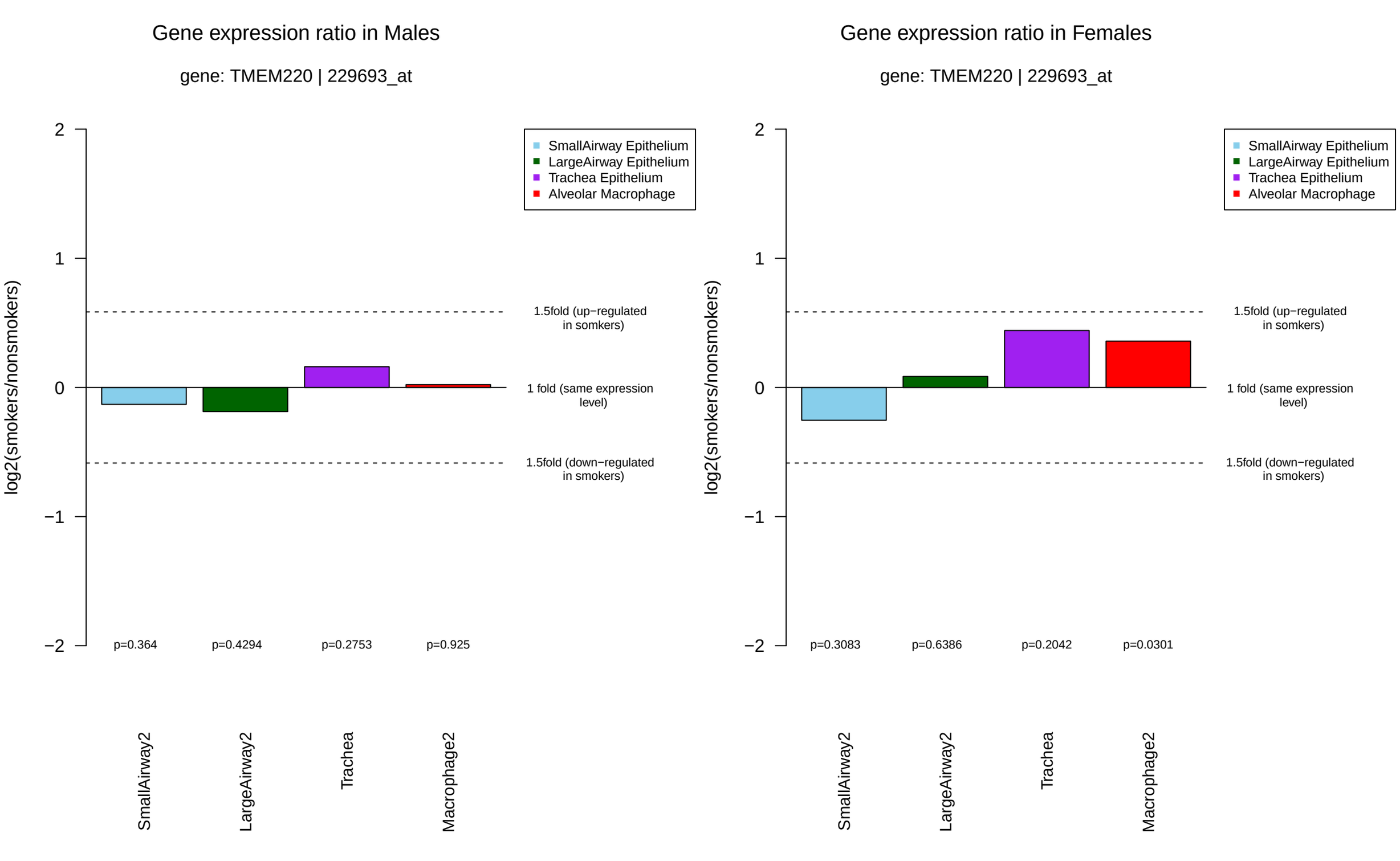
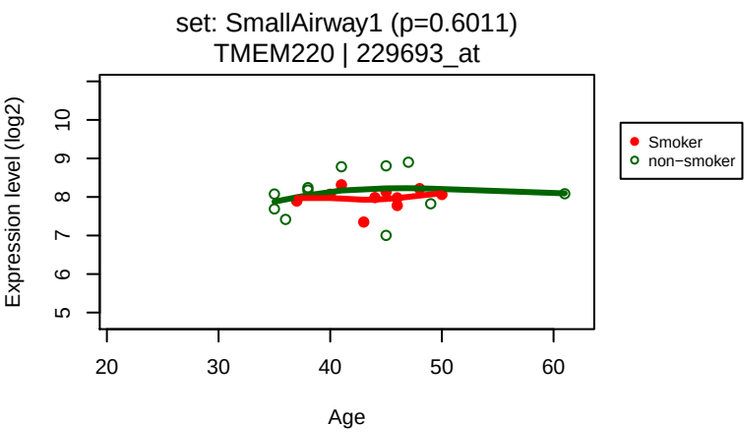
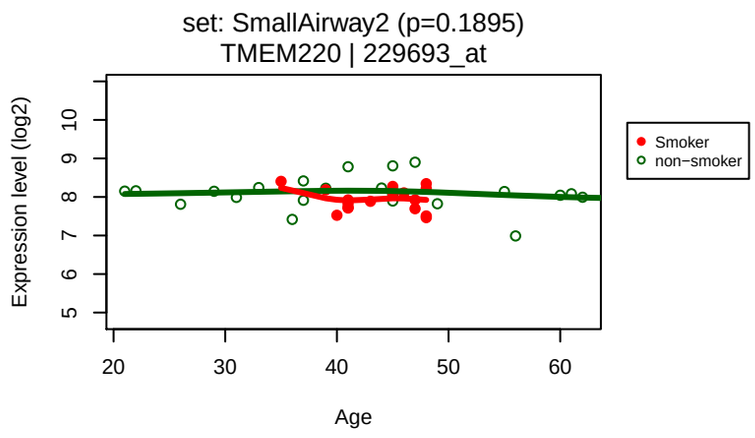
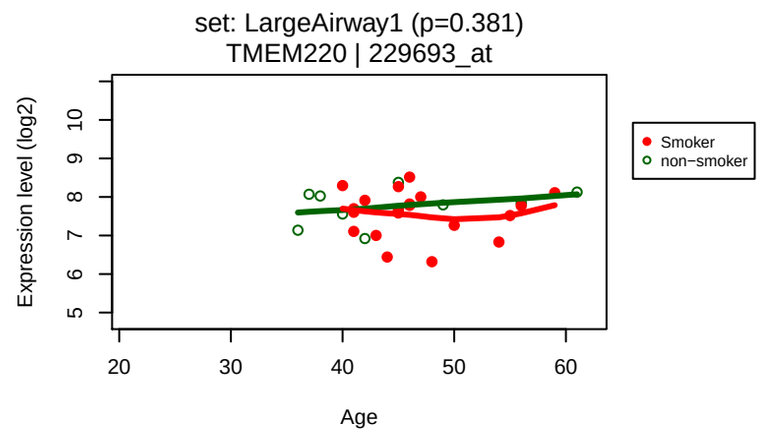

Gene expression of smokers and nonsmokers
gene: TMEM220 | 229693_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6011
p=0.1895
p=0.6722
p=0.381
p=0.6232
p=0.0402
p=0.1531
p=0.4052
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

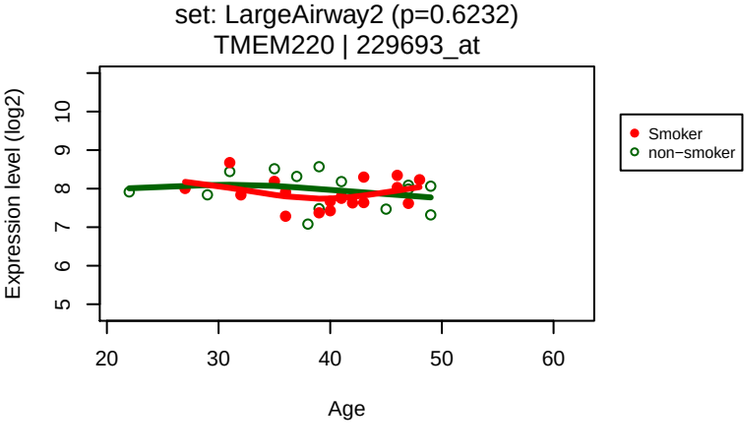

Gene expression ratio (smokers/nonsmokers)
gene: TMEM220 | 229693_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6011
p=0.1895
p=0.6722
p=0.381
p=0.6232
p=0.0402
p=0.1531
p=0.4052
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

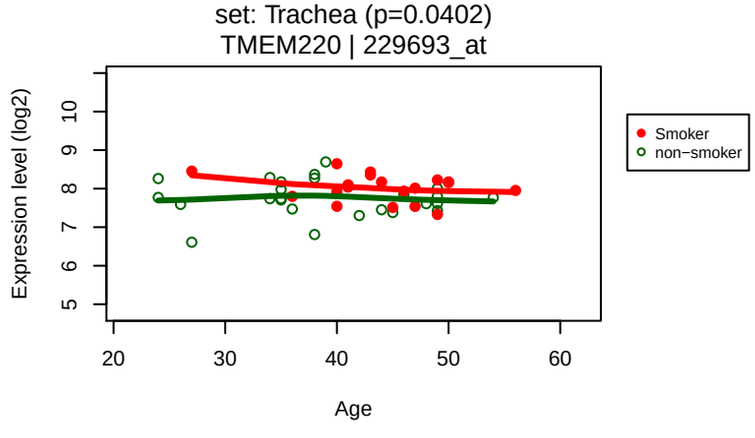

Gene expression ratio in Males
gene: TMEM220 | 229693_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.364
p=0.4294
p=0.2753
p=0.925
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM220 | 229693_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3083
p=0.6386
p=0.2042
p=0.0301
SmallAirway2
LargeAirway2
Trachea
Macrophage2

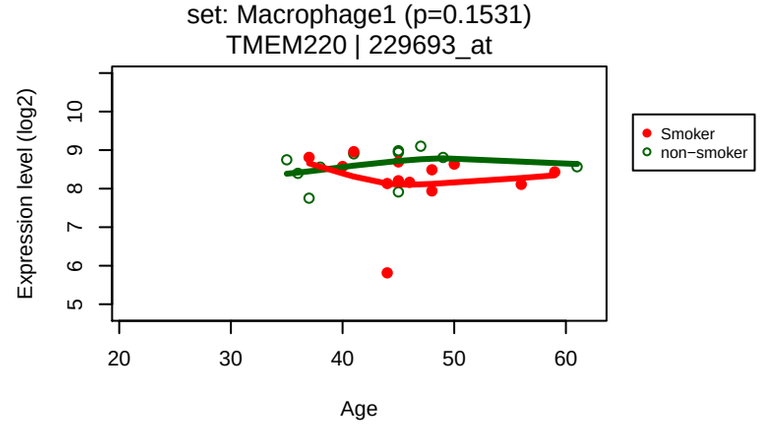

set: SmallAirway1 (p=0.6011)
TMEM220 | 229693_at
Expression level (log2)
Age
Smoker
non−smoker

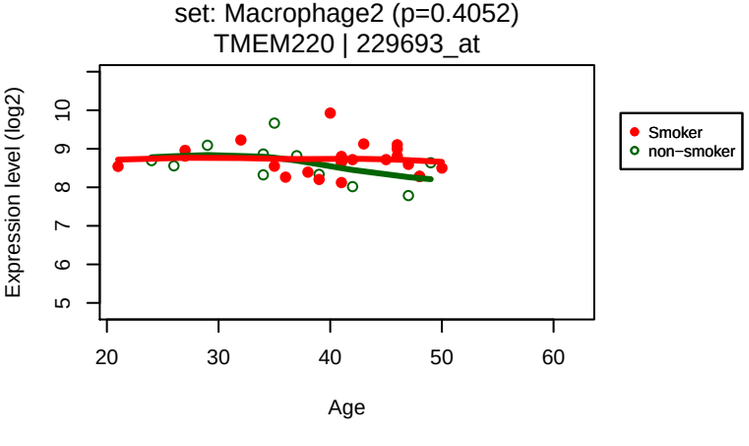

set: SmallAirway2 (p=0.1895)
TMEM220 | 229693_at
Expression level (log2)
Age
Smoker
non−smoker

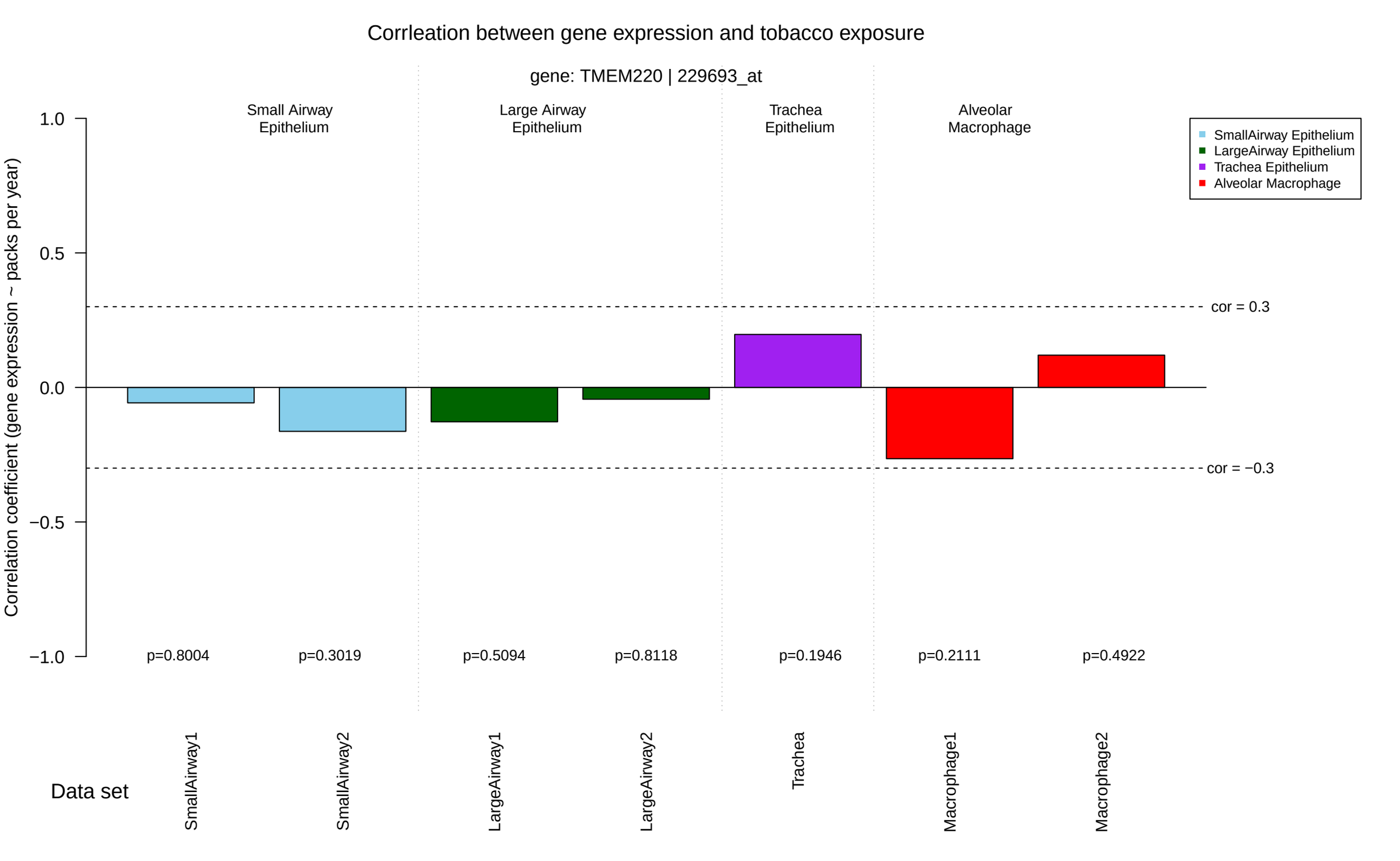

set: LargeAirway1 (p=0.381)
TMEM220 | 229693_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6232)
TMEM220 | 229693_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0402)
TMEM220 | 229693_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1531)
TMEM220 | 229693_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4052)
TMEM220 | 229693_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM220 | 229693_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8004
p=0.3019
p=0.5094
p=0.8118
p=0.1946
p=0.2111
p=0.4922
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2